

# **Black-Box Sparse Adversarial Attack on EEG Recognition**

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# EEG Classification Model and Security Threat

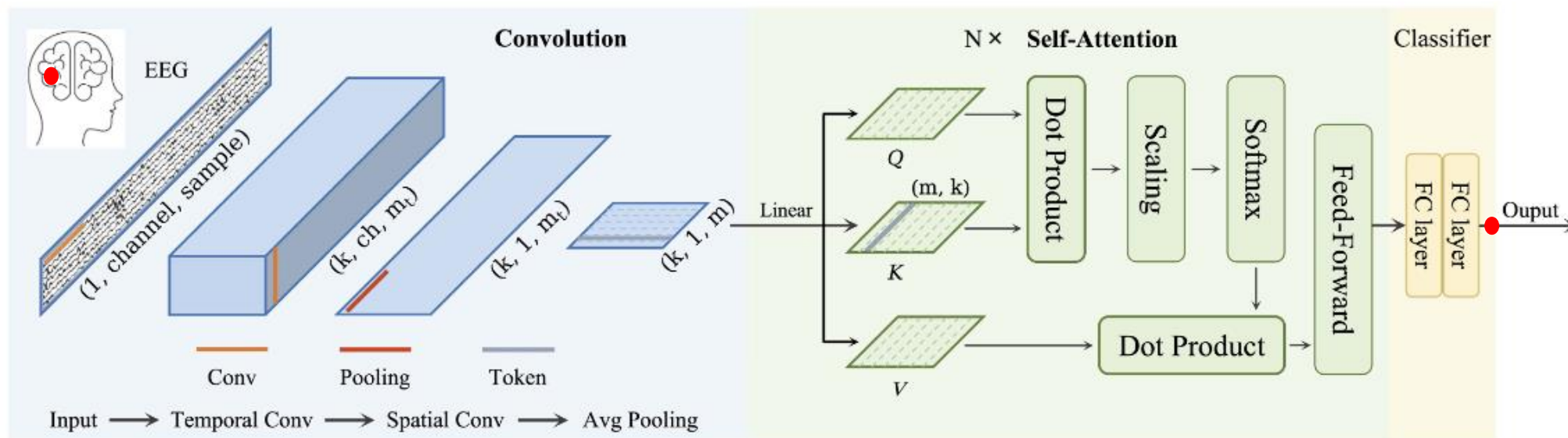


Fig. 1. The framework of Convolutional Transformer (Conformer), including a convolution module, a self-attention module, and a classifier module.

Blackbox Sparse Adversarial Attack

Mô hình/ task

Identification/ dataset=> channel/size of dataset. Time series. Set motion.

Training/testing (9 người)/generalize

EEG Conformer: Convolutional Transformer for EEG Decoding and Visualization, IEEE TRANSACTIONS ON NEURAL SYSTEMS AND REHABILITATION ENGINEERING, VOL. 31, 2023 (classification is being used here as a surrogate target to study vulnerability before moving to the authentication setting)

# EEG Classification Model and Security Threat

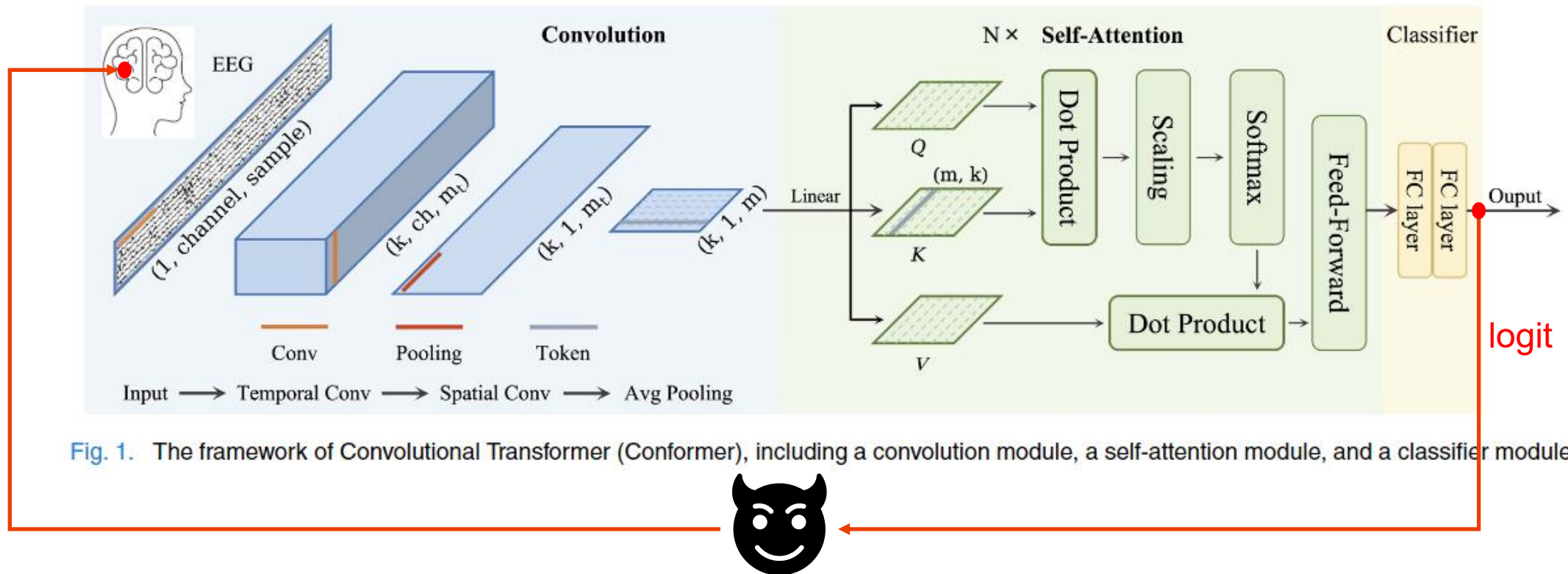


Fig. 1. The framework of Convolutional Transformer (Conformer), including a convolution module, a self-attention module, and a classifier module.



## Blackbox Sparse Adversarial Attack

Can an unauthorized user bypass EEG authentication through a black-box attack?

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# Blackbox Sparse Adversarial Attack

What adversarial attack?

Sparse adversarial attack? Add perturbation trong 1 lượng channel/time/frequency => đầu ra sai (untargeted attack)

=> asr

=> black/white box (threat model/adversarial knowledge model () /data)

=> Ko model/ko data/ chỉ add input logit trước softmax

=> Min number of channel/ which channel to add perturbation to mislead the model

=> For each selected channel, eeg sample divided into x sample in t time.

=> Stealthy sparse (control added perturbation amplitude/ maximum amplitude < 10%, power=> square amp, fixed amplitude over sample)

# Blackbox Sparse Adversarial Attack

⇒  $f(X)$ : mapping from  $X$  to predicted class

⇒ Optimization variable  $\mathbf{E} = [e_{c,t}] \in \mathbb{R}^{C \times T}$

$$\mathbf{X}^{\text{adv}} = \mathbf{X} + \mathbf{E},$$

⇒ Min ( $L_0$  norm ( $E$ ) +  $\lambda L_{\infty}$  ( $E$ ))

⇒ s.t.  $f(X) \neq f(X+E)$   
 $X$  in dataset  $\mathcal{X}_{\text{test}}$

$$\hat{y}(\mathbf{X} + \mathbf{E}) \neq y(\mathbf{X})$$

$$e_{c,t} = \begin{cases} \alpha_{c,t}, & (c,t) \in \mathcal{S}, \\ 0, & (c,t) \notin \mathcal{S}, \end{cases}$$

⇒ (subset) 2 phase which channel to attack, which eeg sample in that channel to attack (to shorter)

⇒ Maximize diff theo  $e()$

# Blackbox Sparse Adversarial Attack

⇒ Stealthy sparse (control added perturbation amplitude/ maximum amplitude < 10%, power=> square amp, fixed amplitude over sample)

$$\phi_t(\tau) = g_t(\tau) \cos(2\pi f_t \tau + \psi_t),$$

⇒ Perturbation, apply to eeg signal, we sample this continuous perturbation waveform. Perturbation = discrete signal (with sampling eeg frequency)

⇒ Objective

⇒ (subset ) 2 phase which channel to attack, which eeg sample in that channel to attack (to shorter )

⇒ Maximize diff theo e()

# Blackbox Sparse Adversarial Attack

$$\mathbf{E} = [e_{c,t}] \in \mathbb{R}^{C \times T}$$

- ⇒ How to attack?
- ⇒ (subset ) 2 phase which channel to attack, which eeg sample in that channel to attack (to shorter )
- ⇒ Phase 1: select the subset of channels to attack
- ⇒ Phase 2: For each selected channel in Phase 1, select the subset of samples to attack

# Blackbox Sparse Adversarial Attack

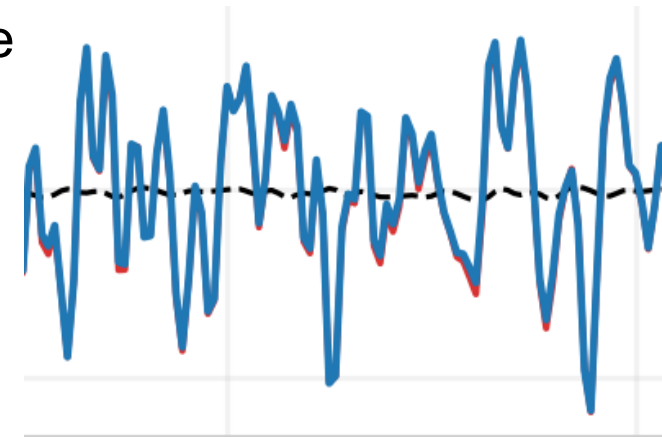
$$\mathbf{E} = [e_{c,t}] \in \mathbb{R}^{C \times T}$$

- ⇒ Define  $h_k(X)$ : mapping from  $X$  to classification score of the class  $k$
- ⇒ Define  $E_c$  in  $\mathbb{R}^{C \times T}$ , but with only channel  $c$
- ⇒ Phase 1 design
  - ⇒ To guide the design, we define a channel-selecting metric  $\gamma_c$
  - ⇒  $E_c = \cos(2\pi f_c t + \text{phase})$  with sampling time  $t$ ,  $f_c \sim U(0, f_{\max})$
  - ⇒  $\gamma_c(E_c) = \max_k (h_k(X + E_c) - h_k(X))$
  - ⇒  $\gamma_{\text{avg}}(E) = \text{avg}(\gamma_c(E_c))$  (sample avg with multiple realization of  $f_c$ )
  - ⇒ Top-K ( $\gamma_{\text{avg}}(E)$ )  $\Rightarrow$  subset of selected channels
- ⇒ Choose the subset of channel base on



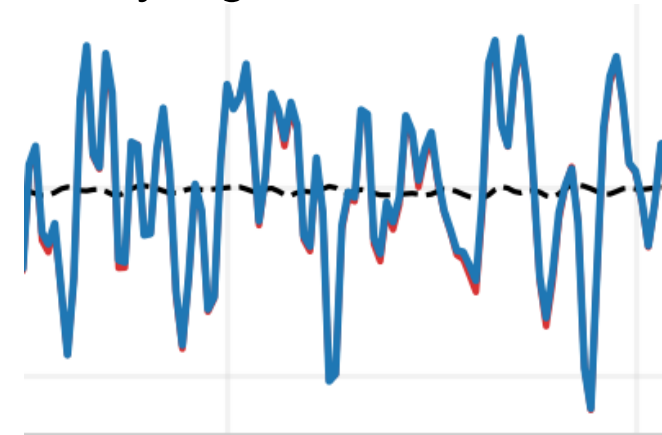
# Blackbox Sparse Adversarial Attack

- ⇒ Define  $h_k(X)$ : mapping from  $X$  to classification score of the class  $k$
- ⇒ Define  $E_{c \setminus \tau}$  in  $\mathbb{R}^{C \times T}$ , but with only channel  $c$  and sample  $\tau$  is non zero
- ⇒ Phase 2 design:
  - ⇒ For each selected channel  $c$
  - ⇒ Sample  $f_{c \setminus \tau_j}$  over  $U(0, f_{\max})$   $J$  times,
  - ⇒ With each sample  $f_{c \setminus \tau_j}$ , we have  $E_{c \setminus \tau_j} = \cos(2\pi f_{c \setminus \tau_j} t + \text{phase})$  with sampling time  $t$ .
  - ⇒  $\gamma_{c \setminus \tau}(E_{c \setminus \tau_j}) = \max_k (h_k(X + E_{c \setminus \tau_j}) - h_k(X))$
  - ⇒  $\gamma_{c \setminus \tau}(E_{c \setminus \tau}) = 1/J \sum_{j=1}^J (\gamma_{c \setminus \tau}(E_{c \setminus \tau_j}))$
- ⇒ Top-x ( $\gamma(E_{c \setminus \tau})$ )  $\Rightarrow$  subset of selected channels sample



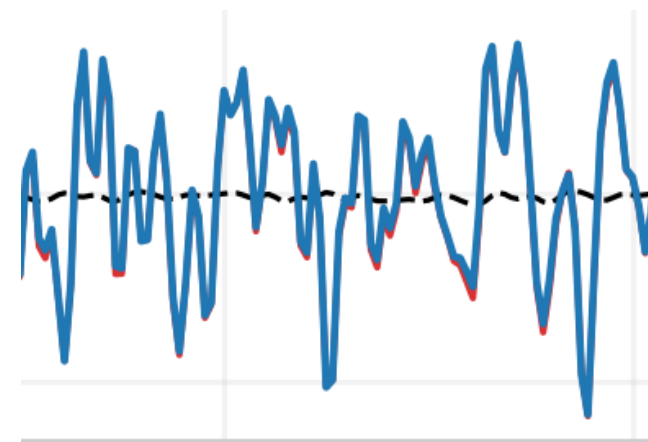
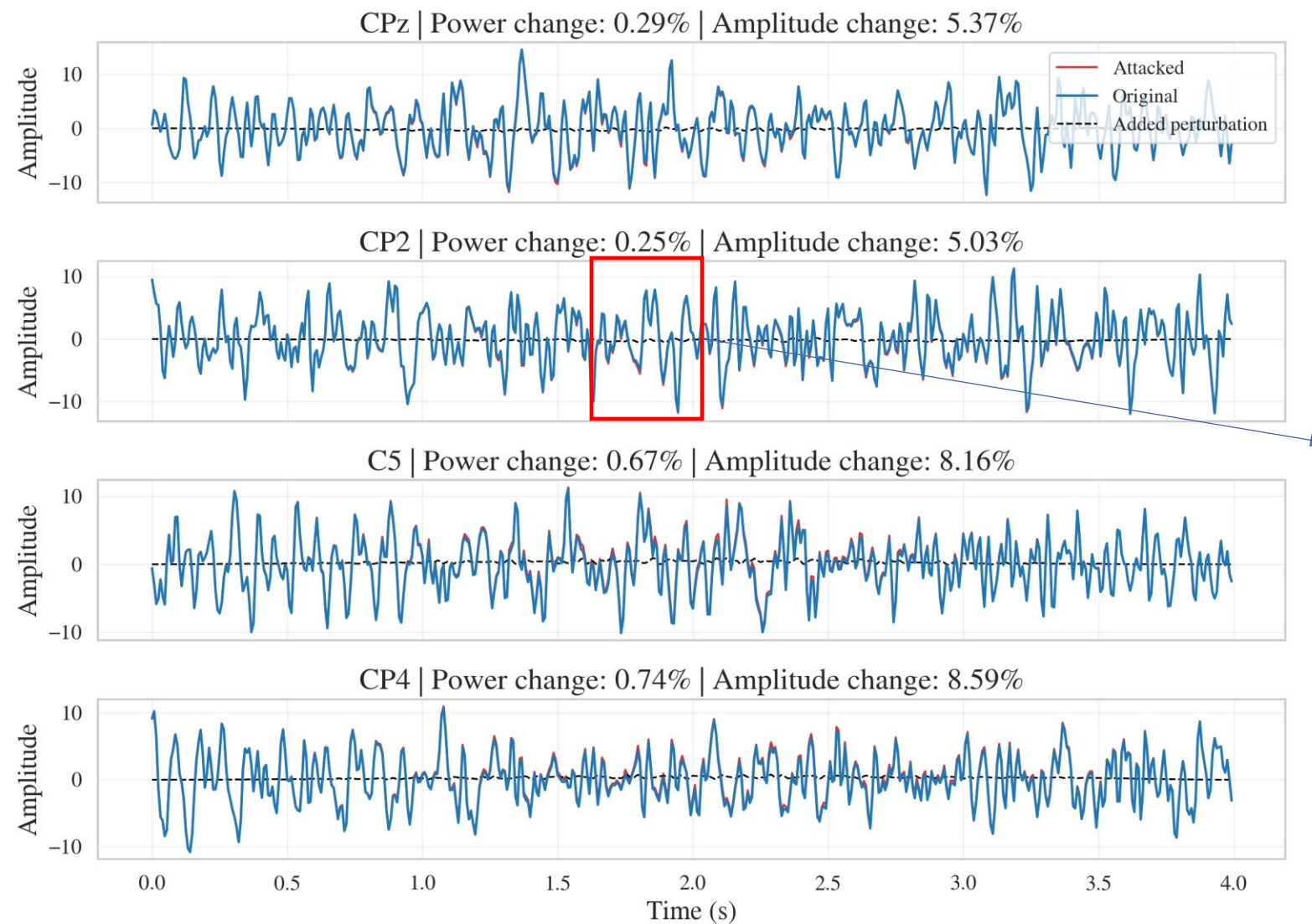
# Blackbox Sparse Adversarial Attack

- ⇒ Define  $h_k(X)$ : mapping from  $X$  to classification score of the class  $k$
- ⇒ Define  $E_{c\backslash\tau}$  in  $R^{C \times T}$ , but with only channel  $c$  and sample  $\tau$  is non-zero
- ⇒ Phase 3 design:
  - ⇒ For subset of selected channels samples
  - ⇒ Determine minimum set of samples on selected channels, to launch the adversarial attack.
  - ⇒ Iterative increase the size of the attacked sample with optimized scaling factor for the perturbation waveform, until the estimated attack success is sufficiently high.



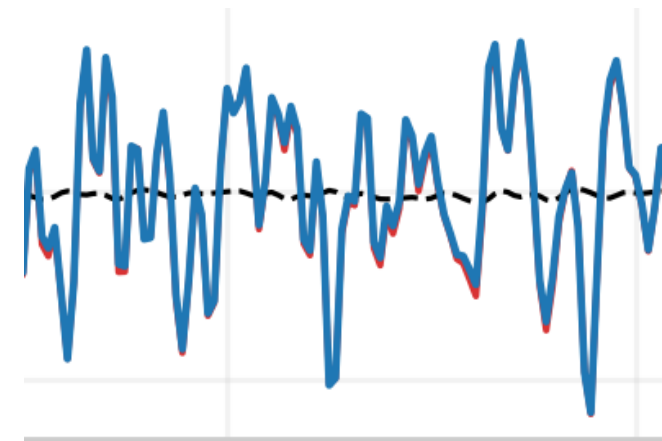
# Result

Sparse Channel Attack Waveforms on Four Attacked Channels (sample 59)

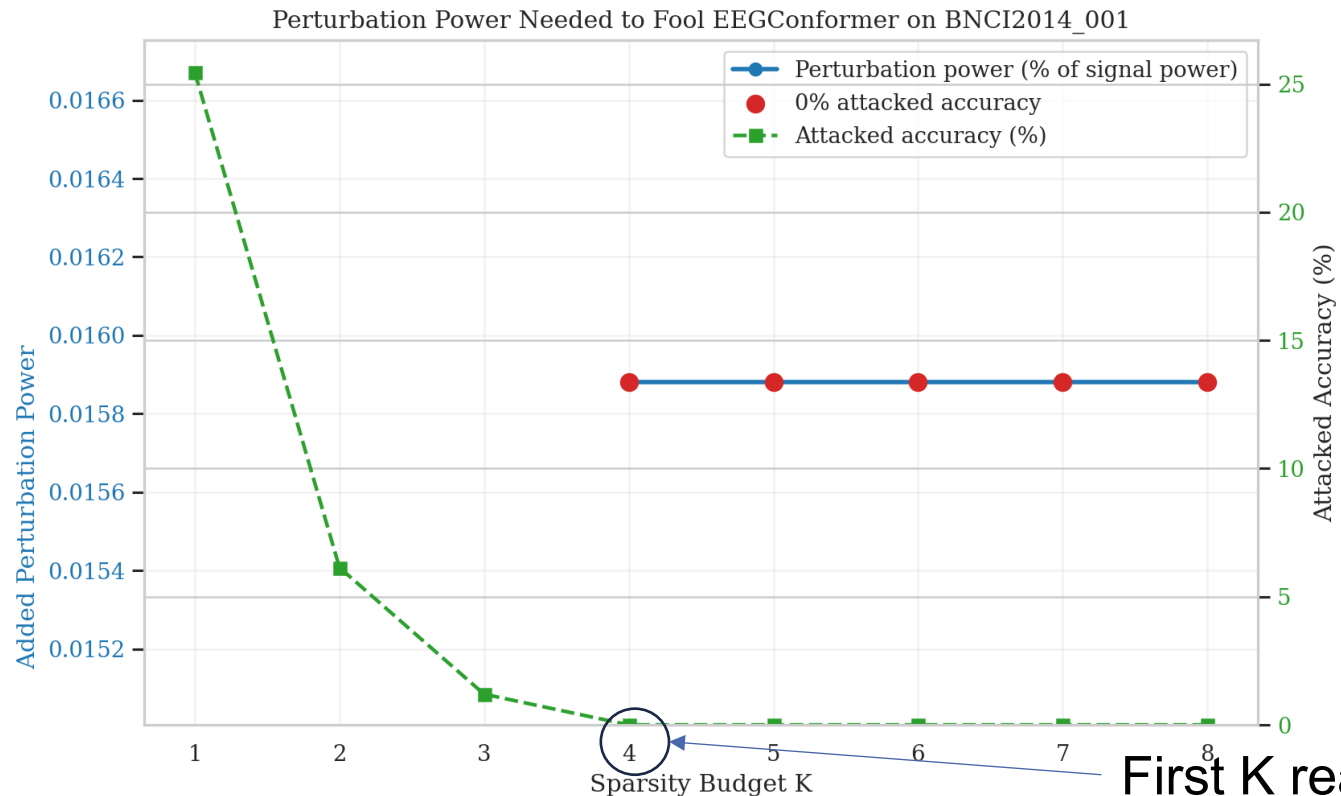


# Blackbox Sparse Adversarial Attack

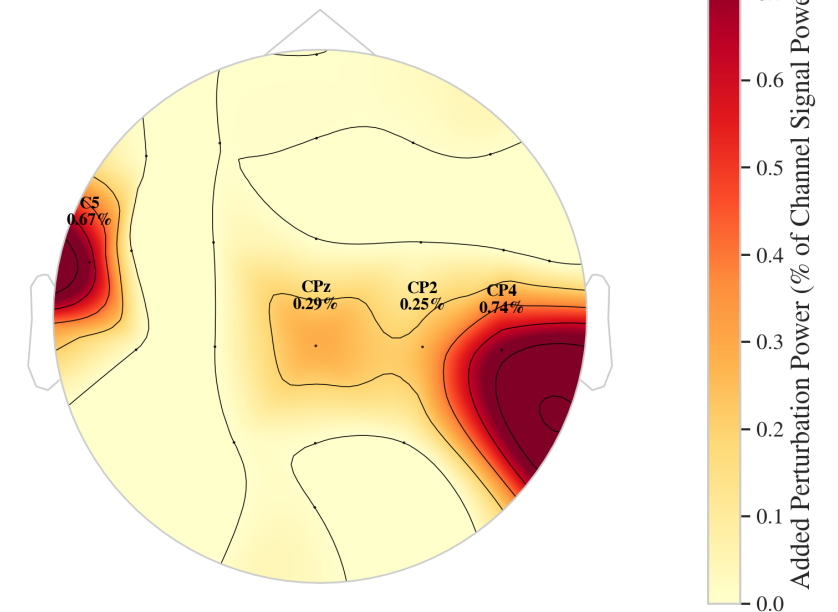
- ⇒ Baseline Filter adversarial attack : whitebox, gradient-based adversarial, all channels and samples
- ⇒ Adversarial design



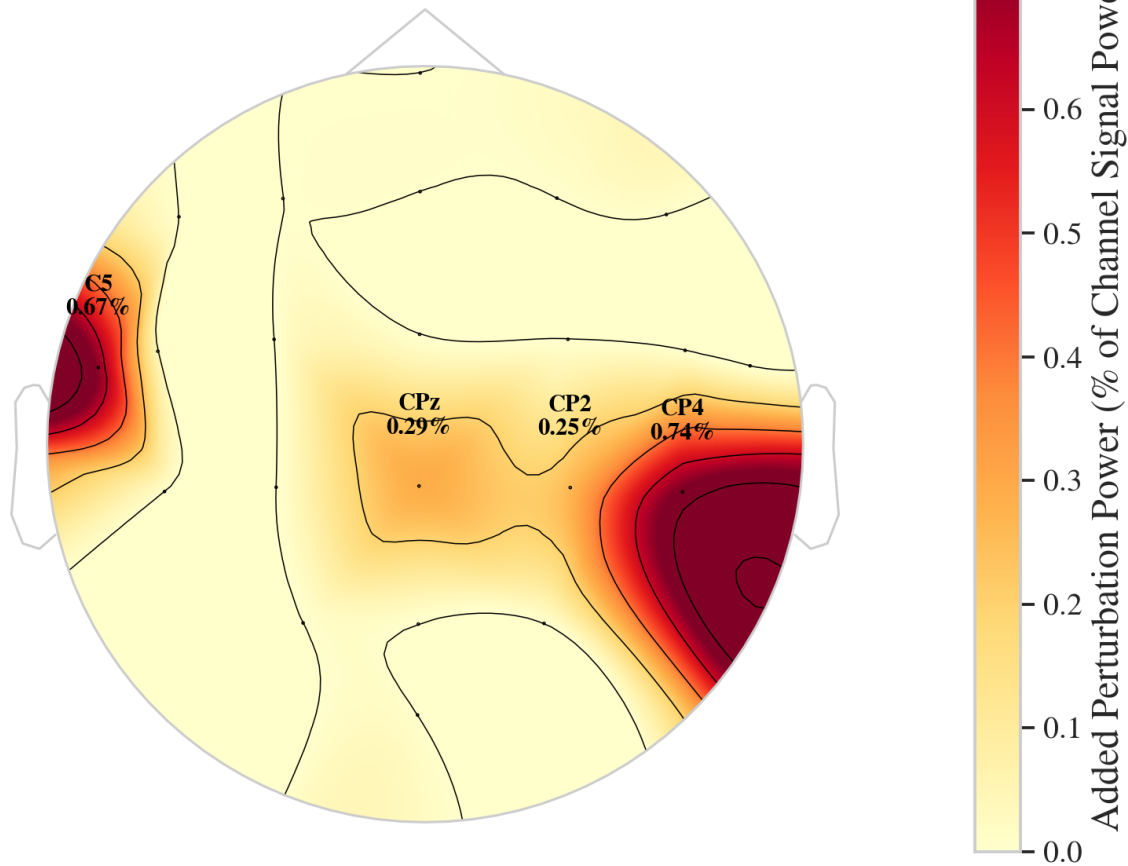
# Blackbox Sparse Adversarial Attack



Sparse Channel Attack: Four Attacked Electrodes

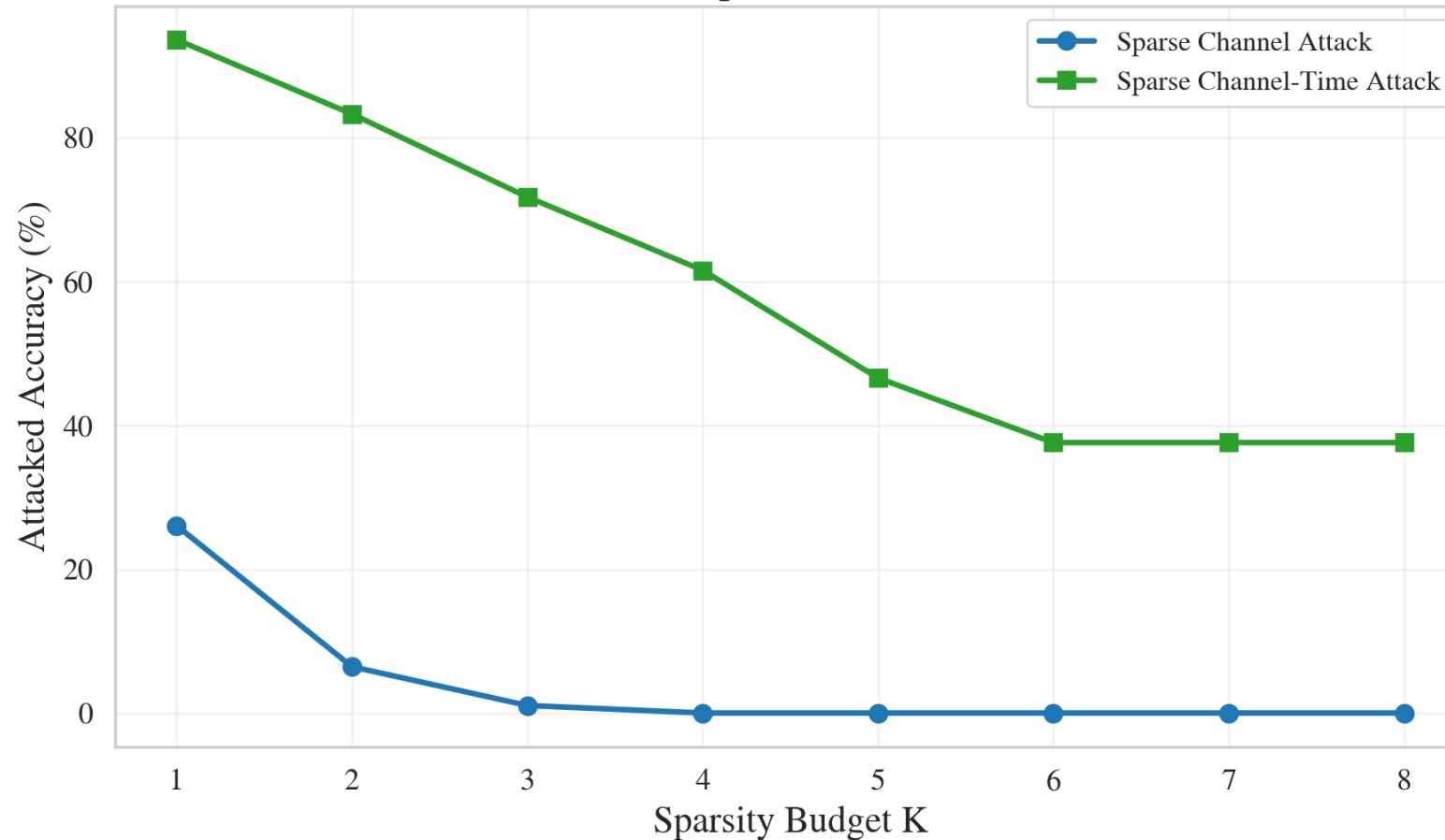


Sparse Channel Attack: Four Attacked Electrodes



The attack reaches 100% success rate with 4 out of 22 channels attacked and at low normalized perturbation power

EEGConformer Under Sparse Attacks on BNCI2014\_001



We also wanted to limit the time and the frequency of the added signal, but so far, it is not as successful.

This is an input-specific attack: it is optimized to fool the model on a particular trial, rather than to generalize across all inputs.

A dataset-level, model-level, or group-level universal attack would be more dangerous, but is also more difficult.

# THANK YOU !



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